

SQL Interview Patterns

Classic coding patterns: duplicates, rankings, running totals, pivots and gaps

SERIES GOAL

Learn the concept → see the pattern → apply it → revise it.

Second-highest salary

One of the classic SQL interview patterns in the source.

Pattern

MAX(salary) WHERE salary < (SELECT MAX(salary) ...)

Concept: subquery + MAX()

Interview tip: explain the idea in words before showing the query.

Find duplicates + delete them safely

Grouping finds them; ROW_NUMBER can help keep one row.

Find duplicates

```
GROUP BY name  
HAVING COUNT(*) > 1
```

Delete duplicates

```
ROW_NUMBER() OVER (PARTITION BY  
email ORDER BY id)
```

Delete rn > 1

Top 3 per department

A pattern that combines window functions and PARTITION BY.

Step 1

ROW_NUMBER() OVER (PARTITION BY dept ORDER BY salary DESC)

Step 2 → filter rn <= 3

The important pattern is “rank inside each group, then filter the rank.”

Running totals without GROUP BY

Window SUM() keeps individual rows visible.

SUM(amount) OVER (ORDER BY order_date)

Each row keeps its detail, while the window function adds the cumulative value.

Pattern: window function + ORDER BY

Nth-highest salary

Correlated subquery pattern from the source notes.

Idea

Count how many distinct salaries are higher than the current salary.

For Nth highest → N-1 higher distinct salaries.

Interview cue: clarify whether ties should count as the same rank.

Rows to columns: PIVOT

A common reporting transformation.

Input

dept | month | revenue

IT | Jan | 100

IT | Feb | 120

Output

dept | Jan | Feb

IT | 100 | 120

Consecutive dates and streaks

The source uses ROW_NUMBER + DATEADD to create grouping keys.

Pattern

date - ROW_NUMBER() → grouping key

Rows sharing the same derived key belong to the same consecutive “island”.

This is the classic “islands and gaps” pattern for streak problems.

How to solve unfamiliar SQL questions

Use a repeatable reasoning process.

Clarify the output

Identify grain

Choose joins

Filter rows

Group / window

Validate edge cases

Found this helpful?

Help others learn too!

Save

Follow

Share

Comment

@VenuMaheshuni

Follow for more Data Engineering content

[linkedin.com/in/venumaheshuni](https://www.linkedin.com/in/venumaheshuni)

Save this carousel for your next project